Pseudocode

Value Iteration (Deterministic Version)

```
for each s in S, V[s] := 0
t := 0
do
t := t + 1; V_last[s] := V[s]
for each s in S:
    for each a in A:
        Q[s, a] = R[s, a] + γ V_last[s']
    π[s] := argmax_a: Q[s, a]
    V[s] := Q[s, π[s]]
until max_s |V[s] - V_last[s]| < ε
```

Value Iteration (Stochastic Version)

```
for each s in S, V[s] := 0
t := 0
do
t := t + 1; V_last[s] := V[s]
for each s in S:
    for each a in A:
        Q[s, a] = R[s, a] + γ Σ_s' T(s,a,s') V_last[s']
    π[s] := argmax_a: Q[s, a]
    V[s] := Q[s, π[s]]
until max_s |V[s] - V_last[s]| < ε
```

Q-Learning

```
initialize Q[s, a] := 0
for all episodes
    initialize s
    do
        choose a based on Q
        execute a and observe s' and r
        Q[s, a] := (1-α)Q[s, a] + α(r + γ max_a' Q[s',a'])
        s := s'
    while not terminal?( s)
```

Sarsa

```
initialize Q[s, a] :=0
for all episodes
    initialize s and pick a
    do
        execute a and observe s' and r
        pick a'
        Q[s, a] := (1-α)Q[s, a] + α(r + γ Q[s',a'])
        s := s', a := a'
    while not terminal?( s)
```